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Palaeomagnetism of the Pico Formation, Santa Paula Creek, Ventura Basin, California

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SUMMARY

In the western Ventura Basin in southern California, marine turbidites in the Pico Formation are exposed along Santa Paula Creek in a southeast-dipping homocline. The section is about 3 km thick and contains normal and reverse polarity that was acquired before the development of the homocline. Although the sediment contains a secondary component of normal polarity that cannot be removed entirely by thermal or alternating-field demagnetization, the mean declination for 26 sites (12 normal polarity, 14 reverse polarity) shows a clockwise rotation of 19.8° ± 9.0° that is similar to the rotation recorded in the Saugus Formation, a coeval, non-marine deposit in the eastern Ventura Basin that was studied by Levi *et al.* (1986). The partial overprint of the secondary component of normal polarity results in a mean inclination for each polarity that does not match the inclination of an axial dipole field (the inclination is too steep in normal polarity and too shallow in reverse polarity); still, the inclination for the combined sites is close to the expected inclination (52.0° versus 53.9°). Overall, turbidites in the Pico Formation are suitable for a magnetostratigraphic investigation, but caution is required if the palaeomagnetic data are to be used in a tectonic study.

Key words: California, palaeomagnetism, Pico Formation, Santa Paula Creek, Ventura Basin.

INTRODUCTION

It has been hypothesized and supported by palaeomagnetic data that motion between the North American and Pacific plates produced clockwise rotation in tectonic blocks bounded by pre-existing northeast-to-north-trending normal faults in southern California (Kamerling & Luyendyk 1979; Luyendyk, Kamerling & Terres 1980; Hornafius 1985; Kamerling & Luyendyk 1985; Luyendyk *et al.* 1985; Terres & Luyendyk 1985; Hornafius *et al.* 1986; Liddicoat 1988, 1990). In that model, the rotation occurred about vertical axes and was accommodated by left-lateral displacement on the defining faults (Luyendyk *et al.* 1980). The palaeomagnetic data show that rotation of the Western Transverse Block—the region between the Santa Ynez, San Gabriel and Dume faults (Luyendyk *et al.* 1985) (Fig. 1)—has approached 90°, and that the motion began no earlier than the Miocene (Liddicoat 1990) and continued into the Brunhes Normal Chron (Levi *et al.* 1986). Jackson & Molnar (1990) summarize the structural and seismic data that help to establish the source of the deformation, and Luyendyk (1991) provides a model for the possible cause of the two major episodes of crustal rotation in southern California since the early Miocene.

The palaeomagnetic evidence for the most recent rotation in the Western Transverse Block is recorded in the non-marine Saugus Formation that is exposed between the Santa Ynez Fault and the Oak Ridge Fault on the eastern side of the Ventura Basin (Levi *et al.* 1986). Part of the Pico Formation, a marine turbidite sequence exposed on the western side of the basin, is bracketed by those faults and should reflect the easterly declination in the Saugus Formation if the two formations are in the same tectonic block. That possibility is examined in this report, as well as the suitability of marine turbidites for documenting palaeomagnetic polarity and tectonic rotation.

LOCALITY, STRATIGRAPHY AND LABORATORY PROCEDURES

The Pico Formation is exposed along Santa Paula Creek (Fig. 2) where Natland & Kuenen (1951) conducted a pioneer study of marine turbidites. The strata uniformly

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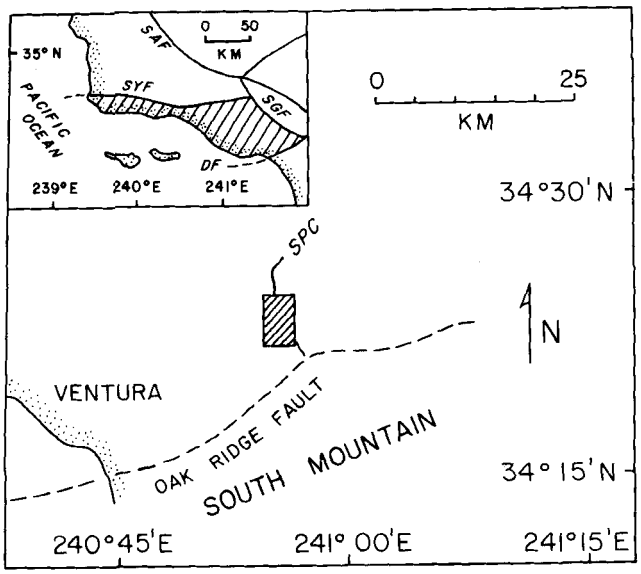


Figure 1. Map of the central Ventura Basin showing the location of Santa Paula Creek (SPC) that flows through the study area (shaded rectangle) north of the Oak Ridge Fault (dashed line). The Saugus Formation was studied by Levi *et al.* (1986) about 40 km east of Santa Paula Creek. Insert map shows the Western Transverse Block (shaded) that is bordered by the Santa Ynez Fault (SYF), San Gabriel Fault (SGF), and Dume Fault (DF). The Western Transverse Block is defined by Luyendyk *et al.* (1985), and the fault pattern is from Jennings (1973).

dip 70° to the SSE (160°) and are approximately midway in a 6 km section of sedimentary rocks assigned to the Quaternary. Interpreting from foraminiferal and molluscan ecology, regional palaeogeographic reconstructions, and analogies with present offshore topography that borders southern California, the bottom environment between turbidity currents was still water greater than 270 m deep (Crowell *et al.* 1966).

The Santa Paula Creek section is about 3 km thick and consists of interbedded mudstone, siltstone and coarse- to medium-grained sandstone; individual beds vary in thickness from several metres to less than a centimetre. Sedimentary structures typical of turbidites [graded bedding, load casts, flame structures, overfolds, rip ups, pull aparts, and various sole marks (Crowell *et al.* 1966)] are abundant and easy to recognize.

Although indurated, the sediment is too soft to core with a portable rock drill cooled with water. Instead, oriented hand samples that could be cut in the laboratory into subsamples (cubes 2.54 cm on a side) were taken from the bank of the creek or from a road excavation. The cutting was done on a lapidary saw cooled with kerosene, which does not dissolve the sediment. The combined error during collection of hand samples and laboratory preparation of subsamples is estimated at ±2° in the horizontal and vertical planes. The same error applies to the measurement of the attitude of individual beds because bedding planes are very well defined. There are at least six subsamples for all but two of the sites, and never less than five (Table 1).

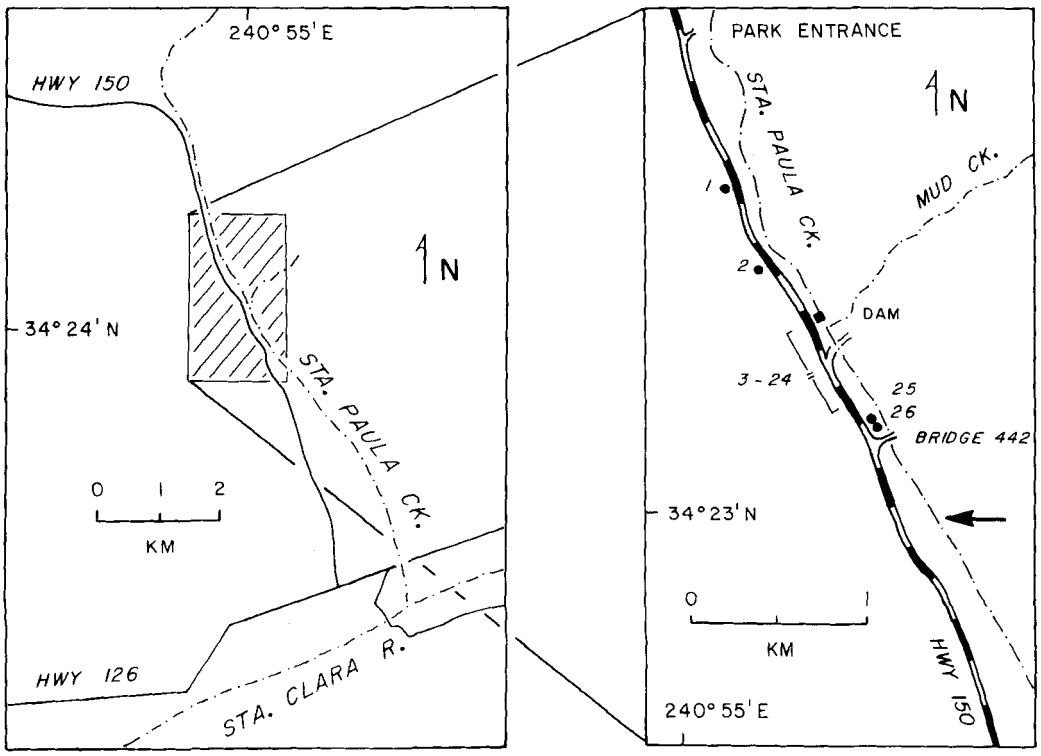


Figure 2. Location map (left) of the study area (shaded) where the Pico Formation is exposed along Santa Paula Creek. Map on the right shows the position of the palaeomagnetic sites. Sites 1 and 2 are on the west side of California Highway 150 northwest of the dam at the gauging station (black square) where Mud Creek enters Santa Paula Creek. Sites 3–26 are between the dam and Bridge 442 that crosses the creek. The black arrow is where R. S. Yeats (personal communication 1991) places an age of 1.0 million years on the Santa Paula Creek section. The position of the sites is given in Table 1.

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Table 1. Palaeomagnetic data for the Pico Formation, Santa Paula Creek, Ventura County.

NUMBER	POSITION	INCL	DECL	N	k	α_{95}	PLAT	PLONG	A_{95}
1	+800	-44.9	190.3	6	297.3	3.9	78.2	19.5	3.2
2	+320	-34.2	195.8	6	261.6	4.2	69.0	24.8	4.0
3	-21	68.2	51.5	6	113.1	6.3	49.9	299.4	9.8
4	-26	60.0	32.2	6	608.3	2.7	63.9	317.1	4.0
5	-31	62.2	10.1	6	156.3	5.4	77.6	287.6	6.8
6	-35	66.0	25.3	6	103.1	6.6	66.0	296.3	10.1
7	-40	67.0	20.4	6	190.8	4.9	68.6	288.4	7.4
8	-47	50.3	15.4	6	12.8	19.5	75.6	341.4	17.7
9	-52	68.3	59.6	5	179.3	5.7	44.4	299.4	8.9
10	-55	61.1	46.3	6	28.3	12.8	53.9	310.7	16.9
11	-58	43.9	42.6	5	113.7	7.2	52.5	342.7	8.5
12	-90	58.3	35.9	5	57.1	10.2	61.7	319.8	13.4
13	-155	61.1	50.3	9	54.2	7.1	50.8	313.1	9.8
14	-220	-54.3	186.1	6	46.7	9.9	85.3	332.0	13.5
15	-227	-41.2	182.7	6	154.2	5.4	79.0	57.8	6.2
16	-234	-39.8	190.4	12	180.2	3.2	75.2	30.4	3.4
17	-237	-47.9	200.8	6	305.6	3.8	71.5	352.1	3.9
18	-270	-38.4	194.3	6	122.1	6.1	72.2	22.2	4.8
19	-305	-35.6	197.9	6	175.7	5.1	68.5	19.1	4.5
20	-355	62.4	76.9	6	200.1	4.7	31.8	306.5	6.5
21	-357	-37.8	193.2	6	402.5	3.3	72.5	26.3	2.9
22	-377	-44.3	182.3	6	435.7	3.2	81.5	57.0	2.9
23	-395	-36.1	187.4	6	143.7	5.6	74.3	44.6	5.5
24	-430	-49.7	182.8	6	155.7	5.4	85.6	40.4	6.7
25	-650	-41.2	178.3	7	117.6	5.6	79.2	80.5	6.3
26	-680	-41.2	197.7	7	957.7	2.0	71.3	11.2	2.0

NUMBER: Site number. POSITION: Metres north (+) or south (−) of the dam in Santa Paula Creek. INCL: Mean inclination (degrees) in stratigraphic coordinates of best-fit directions after thermal demagnetization. DECL: Mean declination (degrees) in stratigraphic coordinates of best-fit directions after thermal demagnetization. N: Number of subsamples measured. k: Fisher precision parameter. α_{95}: Alpha-95: radius (degrees) of circle of 95 per cent confidence about the mean palaeomagnetic directon. PLAT: North latitude of the Virtual Geomagnetic Pole (VGP). PLONG: East longitude of the VGP. A_{95}: Alpha-95: radius (degrees) of circle of 95 pr cent confidence about the mean VGP.

Palaeomagnetic measurements were made in a cryogenic magnetometer in a shielded room following demagnetization in Schonstedt equipment. Each subsample was demagnetized at temperature intervals of 50 °C or less above 100 °C, and at some sites a companion suite of subsamples was demagnetized in an alternating field. Even though the palaeomagnetic directions are similar for both methods of demagnetization (Fig. 3), only thermal demagnetization was applied to each subsample at all of the sites. The plotted and tabulated results are from a best-fit, principal component analysis of the data (Kirschvink 1980). The minerals that carry the magnetization that pre-dates the regional folding have blocking temperatures of up to 450 °C (Fig. 4).

PALAEOMAGNETIC POLARITY

The Santa Paula Creek section contains normal and reverse polarity as shown in Fig. 5. Reverse polarity occurs at sites 1

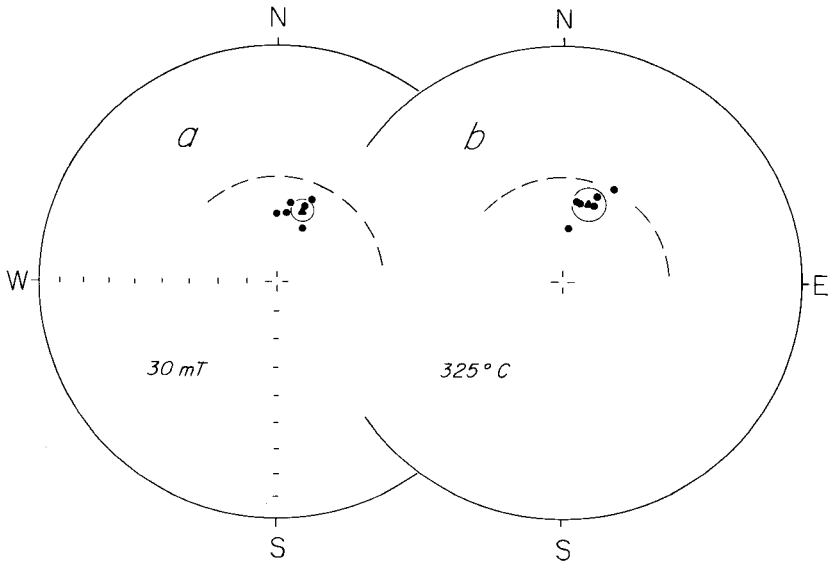


Figure 3. Equal-area plot of palaeomagnetic directions and circles of 95 per cent confidence for suites of six subsamples that were subjected to (a) alternating-field demagnetization at 30 millitesla (mT), and (b) thermal demagnetization at 325 °C. Each method of demagnetization produces directions with a declination that is rotated clockwise and an inclination that is steeper than the inclination of an axial dipole field (53.9°; dashed circle). Data are plotted on the lower hemisphere.

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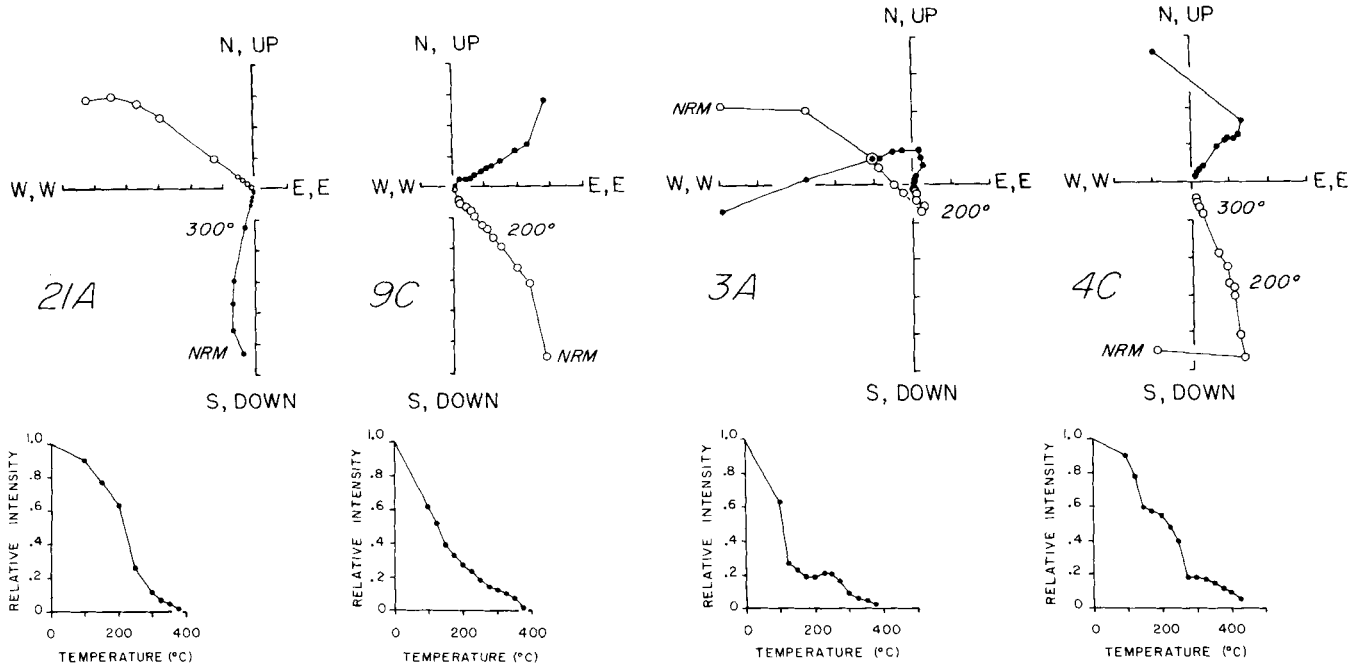
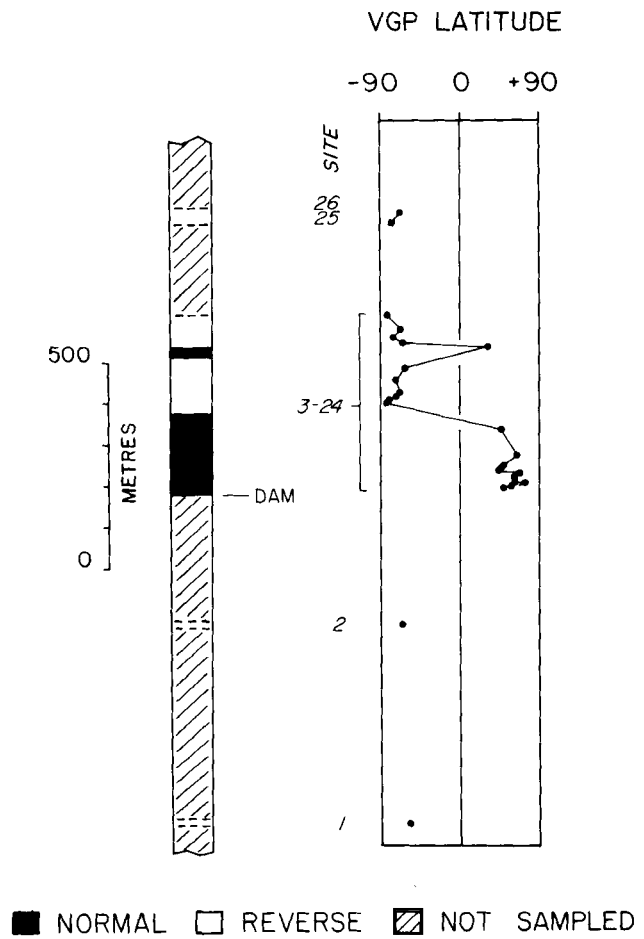


Figure 4. Vector component diagrams and normalized intensity plots for subsamples from sites 3, 4, 9 and 21 following thermal demagnetization and a bedding correction (stratigraphic coordinates). Sites 3, 4 and 9 record normal polarity; site 21 records reverse polarity. The subsamples have a blocking temperature up to about 450 °C. In the vector component plots, solid circles are projections on the NS–EW plane and open circles are projections on the EW–vertical plane. Divisions on the axes are 1×10^{-3} Am^{-1}.



and 2, which are 800 and 320 m northwest (upstream) of the dam at a gauging station on the creek; the sites are in the road excavation on the west side of California Highway 150 that parallels the creek (Fig. 2). The remaining sites are in the bank of the creek where sites 3–13 [21–155 m southeast (downstream) from the dam] have normal polarity. Other than site 20 (355 m downstream from the dam), all of the other sites have reverse polarity. The uppermost one (site 26) is 680 m downstream from the dam and 120 m northwest of Bridge 442 that crosses the creek (Fig. 2). The contact between the Pico Formation and overlying Santa Barbara Formation is about 200 m south of site 26.

The subsamples at each site of normal polarity except site 9 contain an overprint of reverse polarity that predates the most recent folding. A subsample at site 12 is distinctive in that the normal and reverse components are nearly antipolar and are rotated about 30° clockwise (Fig. 6).

The biostratigraphy of the Santa Paula Creek section places a Pliocene age on the Pico Formation (Crowell *et al.* 1966). Because there are no volcanic ash layers in the section to correlate with the tephrochronology for southern California (Sarna-Wojcicki *et al.* 1984; 1991), it is not possible to match with certainty the polarity zones to a reversal time-scale (Mankinen & Dalrymple 1979). Fig. 7 shows three possibilities for the age of the zones where it is assumed that the 3 km Santa Paula Creek section spans the

Figure 5. Palaeomagnetic polarity in the section of the Pico Formation sampled along Santa Paula Creek. The position of the sites is given in Table 1. The solid circles are the latitude of the Virtual Geomagnetic Poles for the site mean directions.

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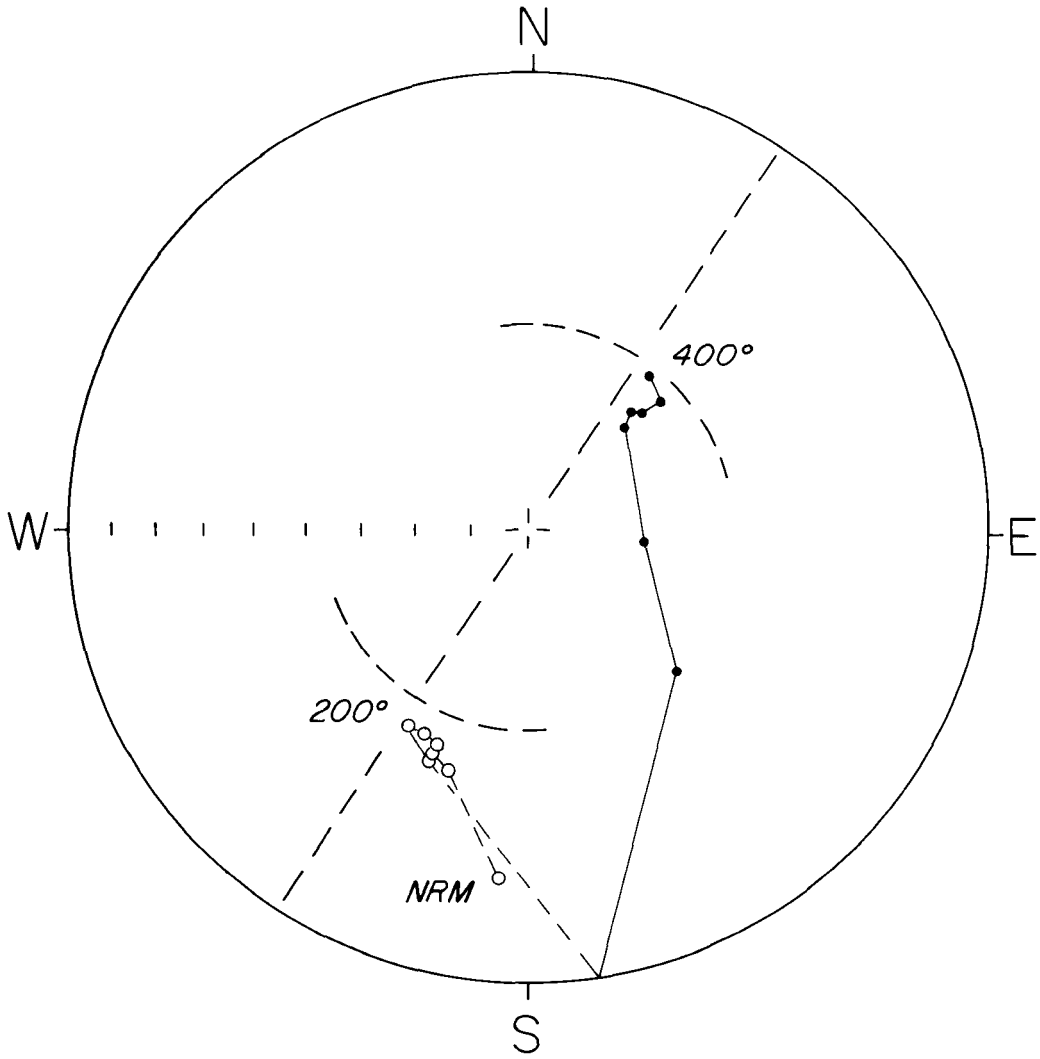


Figure 6. Equal-area plot of palaeomagnetic directions in stratigraphic coordinates during step-wise thermal demagnetization to 400 °C for subsample 12A from site 12 (normal polarity). Note the presence of both polarities, each rotated clockwise by about 30° (dashed line), and the steep and shallow inclination for normal and reverse polarity, respectively; rotation is larger for normal polarity. Solid circles are plotted on the lower hemisphere, open circles are plotted on the upper hemisphere.

Pliocene as presently defined, an interval of about 3.0 Myr. The 134 m of normal polarity exposed in the stream-bed between sites 3 and 13 are about 125 m of vertical section when the 70° SSE dip of the bedding is removed. On the basis of a uniform sedimentation rate for the 3 km section, the overall sedimentation rate is about 1 mm yr^{-1}, and the 125 m represent about 125 000 yr. The 1 mm sedimentation rate corresponds favourably with the minimum rate calculated for Pliocene–Pleistocene sediments in the western part of the Ventura Basin where the magnetic polarity was measured in a core from the Saticoy Oil Field (Blackie & Yeats 1976). Although the time interval is nearly a third shorter than the Olduvai Normal Subchron (1.87–1.67 Ma BP), which is the longest subchron in the Matuyama Reverse Chron, equating the two zones is at least a possibility. That is especially so because downsection (upstream) from the dam where the beds are covered by debris in the stream-bed, the length of the zone can be extended by an unknown amount. A similar adjustment is prevented at the upper boundary of the zone because the separation between normal (site 13) and reverse (site 14) polarity was narrowed to about 6 m (about 6000 yr).

Allowing that the Olduvai Normal Subchron is present in the Santa Paula Creek section, the identity of the normal polarity represented by site 20 is speculative because a three-fold reduction in sedimentation rate is needed to place the polarity in the next younger subchron for which there is general acceptance (the Cobb Mountain Normal Subchron at 1.12 ± .02 Ma BP; Mankinen, Donnelly & Grommé 1978). An even slower sedimentation rate is required if the Jaramillo Normal Subchron (0.97–0.90 Ma BP) is assigned to site 20. However, a normal subchron younger than 1.0 Ma BP is inconsistent with an interpolated age of 1.0 Ma BP for strata in the upper part of the Santa Paula Creek section south of Bridge 442 (R. S. Yeats, personal communication 1991); Fig. 2.

Two other less likely [on the basis of the geology in adjacent areas (R. S. Yeats, personal communication 1991)] interpretations of the observed polarity sequence can be offered (columns B and C in Fig. 7). The normal polarity

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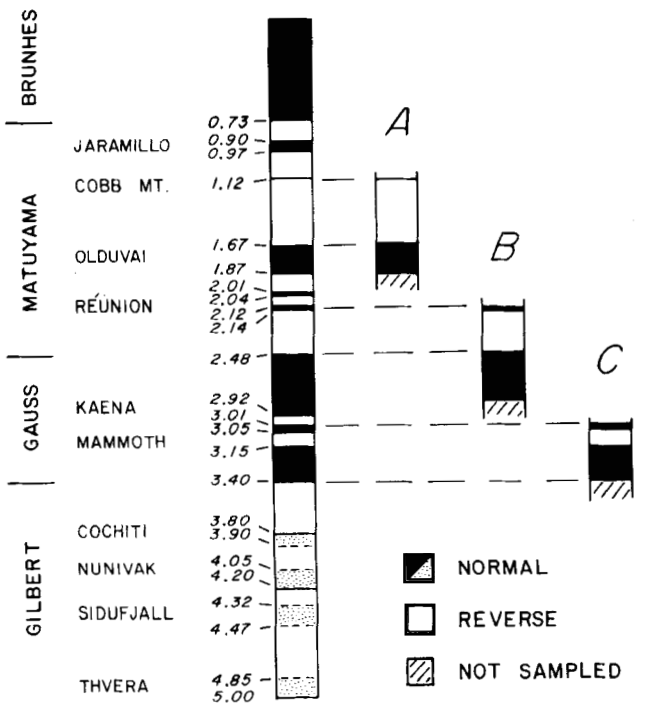


Figure 7. Possible assignments of the palaeomagnetic polarity in the Pico Formation along Santa Paula Creek. The polarity time-scale is from Mankinen & Dalrymple (1979) and the numbers adjacent to the polarity column are millions of years before present.

spanned by sites 3–13 might be the upper half of the Gauss Normal Chron [near the top of the Kaena Reverse Subchron (2.92 Ma BP) to the Gauss/Matuyama boundary (2.48 Ma BP)]. The normal polarity at site 20 could then represent one of the Réunion Normal Subchrons (2.14–2.12 Ma BP), and the underlying reverse polarity (sites 1 and 2) might fall within the Gilbert Reverse Chron (>3.4 Ma BP) or perhaps in the Kaena (3.01–2.92 Ma BP) or Mammoth (3.15–3.05 Ma BP) reverse subchrons in the Gauss Normal Chron. Or, sites 3–24 are the lowermost Gauss Normal Chron (<3.4 Ma BP), and site 20 separates the Mammoth and Kaena reverse subchrons; sites 25 and 26 would be in the Matuyama Reverse Chron (<2.48 Ma BP). Unfortunately, the absence of the Bailey Ash (1.2 ± 0.2 Ma; Izett, Naeser & Obradovich 1974), which is in the Saticoy Oil Field core studied by Blackie & Yeats (1976), in the Santa Paula Creek section prevents a palaeomagnetic correlation of the magnetostratigraphy at these localities.

DISCUSSION

Regardless of the interpretation placed on the polarity pattern in the Santa Paula Creek section, the mean declination for all sites records a clockwise rotation of 19.8° ± 9.0° when reverse declination is inverted through the origin (Fig. 8, Table 2). The inferred rotation is nearly four

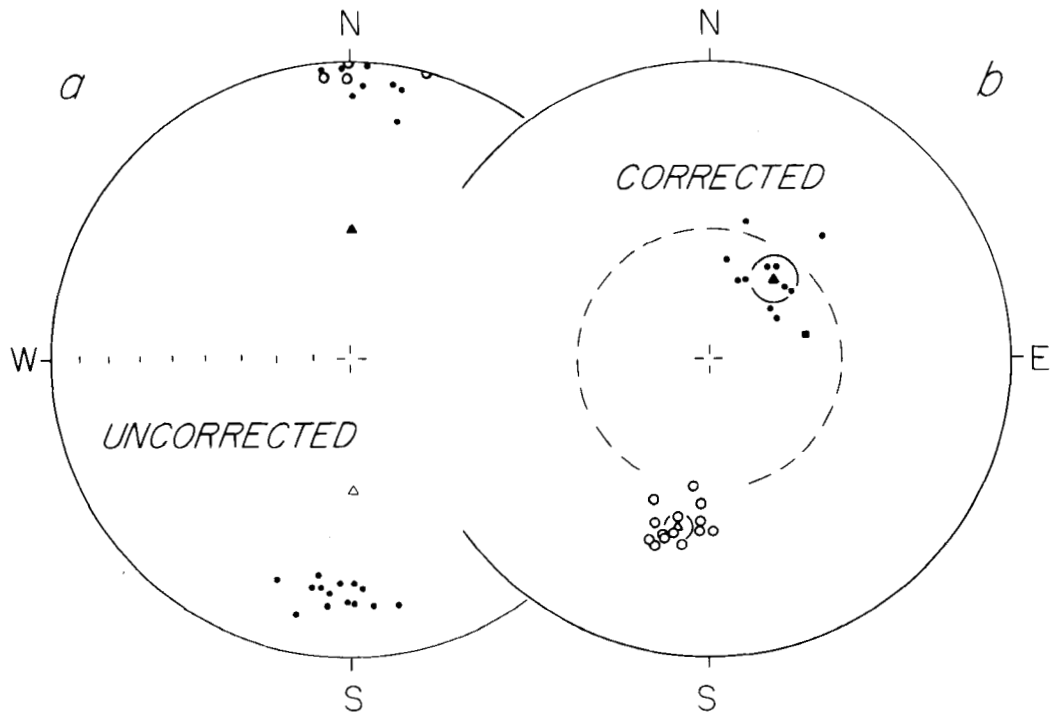


Figure 8. Equal-area plot of site mean palaeomagnetic directions (Table 1) (a) before, and (b) after correction for the attitude of the beds in Santa Paula Creek. The axial dipole field direction is represented by triangles in the plot of uncorrected directions (geographic coordinates); triangles in the plot of corrected directions (stratigraphic coordinates) are the mean direction with the associated 95 per cent circle of confidence for each polarity. Note the clockwise bias of the data and that the mean inclination for normal polarity is steeper than the inclination of an axial dipole field (53.9°: dashed circle) and shallower for reverse polarity. The solid square at $D = 62.4°$, $I = 76.9°$ is the single horizon of normal polarity at site 20. Solid circles are plotted on the lower hemisphere; open circles are plotted on the upper hemisphere.

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Table 2. Palaeomagnetic data for the Pico Formation, Santa Paula Creek, Ventura County.

SITES	INCL	DECL	N	k	α_{95}	ΔI	ΔD	R	ΔR	F	ΔF	PLAT	PLONG
3–13, 20	62.0	38.0	12	48.2	6.3	6.3	13.5	38.0	11.7	8.0	6.4	59.0	310.2
1, 2, 14–19, 21–26	−42.1	190.2	14	110.7	3.8	3.8	5.1	10.2	6.4	−11.9	5.0	76.9	26.7
1–26	52.0	19.8	26	25.2	5.8	5.8	9.4	19.8	9.0	−2.0	6.2	72.6	334.7

SITES: Site number—3–13 and 20: normal polarity; 1, 2, 14–19 and 21–26: reverse polarity; 1–26: all the sites with reverse polarity inverted through the origin. INCL: Mean inclination (degrees) in stratigraphic coordinates of best-fit directions after thermal demagnetization. DECL: Mean declination (degrees) in stratigraphic coordinates of best-fit directions after thermal demagnetization. N: Number of sites. k: Fisher precision parameter. α_{95}: Alpha-95: radius (degrees) of circle of 95 per cent confidence about the mean palaeomagnetic direction. ΔI: α_{95}. ΔD: $\sin^{-1}(\sin \alpha_{95}/\cos \text{INCL})$. R: Rotation calculated from the equations of Beck (1980) relative to the North American pole for the Pliocene. ΔR: Adjusted uncertainty on R from Beck (1980) using the conversion factor from Demarest (1983); dispersion of poles is assumed to be 5°. F: Flattening of inclination. ΔF: Adjusted uncertainty on F from Beck (1980) using the conversion factor from Demarest (1983); dispersion of poles is assumed to be 5°. PLAT: North latitude of the Virtual Geomagnetic Pole (VGP). PLONG: East longitude of the VGP.

times as large for data of normal polarity as it is for data of reverse polarity (38.0° ± 11.7° versus 10.2° ± 6.4°), a pattern that is consistent with their relative age. Although the same values were not found for the Saugus Formation (the mean declination is about 23°, about 32° for the Matuyama and about 14° for the Brunhes; Levi *et al.* 1986), the trend is similar: an overall clockwise rotation with the older rocks rotated the most.

There is a difference, though, between the palaeomagnetic data for the Pico and Saugus formations. The mean inclination in the Saugus Formation is similar for reverse and normal polarity when reverse inclination is inverted through the origin (Levi *et al.* 1986), but that is not the case for the Pico Formation. Note in Fig. 8 and Table 2 that the average inclination for all sites of reverse polarity is shallow when compared to the inclination of an axial dipole (reverse) field (−42.1° versus −53.9°), whereas sites of normal polarity have an average inclination that is steep by about a similar amount (62.0° versus 53.9°). Because of the attitude of the beds in Santa Paula Creek, the inclination just described would occur if a normal (Brunhes?) component of magnetization was not completely removed during thermal demagnetization. That possibility exists for sites of normal polarity where there is not a linear decay to the origin in a vector component plot (an example is subsample 9C in Fig. 4). However, the overprint does not seem to be present in subsamples of reverse polarity such as 21A in Fig. 4. A partial overprint on either polarity when the bedding dips to the SSE would also bias the declination. In the long zone of normal polarity (sites 3–13) the declination would be deflected to the east to exaggerate the clockwise rotation, but for reverse polarity the declination would be brought towards the south. Assuming that the same, but of opposite direction, bias occurred in both polarities, the overall rotation would approach the mean for all sites (the 19.8° ± 9.0° cited above).

The problem of the possible presence of an erroneous inclination is critical in a tectonic study where the palaeolatitude is sought. If, for example, the youthfulness of the Pico Formation was not known (i.e. it is reasonable to expect there has not been large-scale north–south translation of the Ventura Basin north of the Oak Ridge Fault since the beginning of the Pliocene), and only one polarity had been encountered in the Santa Paula Creek section, a latitudinal movement of about 10° north or south, depending on whether the polarity was reverse or normal, respectively, could be postulated.

Although the problems just discussed signal caution in tectonic applications of a palaeomagnetic investigation using turbidites, the polarity does seem to be correctly recorded in the Pico Formation, which Blackie & Yeats (1976) discovered in their palaeomagnetic study of the Pliocene–Pleistocene section recovered in a core in the western part of the Ventura Basin. The conclusion that turbidites reliably record palaeomagnetic polarity was also reached by Alvarez & Lowrie (1984), who measured the magnetization in white calcarenite turbidites in the upper part of the Scaglia Limestones (Campanian–Palaeocene) at Furlo, Italy. When the palaeomagnetic directions in one densely sampled bed having reverse polarity were compared with the mean directions for each polarity above and below the bed (coverage was about 10 m on each side of the densely sampled bed), there was no significant difference in the mean directions. In the limestone, the magnetization resides in extremely fine-grained magnetite that is believed to have been aligned by Brownian motion in the water-filled interstices in the presence of a magnetic field (Lowrie & Alvarez 1975); during subsequent compaction, the magnetic grains retained the original alignment.

Turbidites in the Pigeon Point Formation (late Cretaceous) and in nearby strata assigned to the Palaeocene were used by Champion, Howell & Grommé (1984) in a palaeomagnetic investigation to establish the position of the Salinian terrane in coastal central California. The most reliable results came from the thin-bedded layers, and a comparison of the palaeomagnetic directions in undisturbed and contorted layers showed that the magnetization is a post-depositional remanent magnetization (Irving & Major 1964). Normal and reverse polarity are recorded, and the difference in the mean inclination between the late Cretaceous and Palaeocene turbidites is about 4°. Because the overall mean inclination is shallow by nearly 20° when compared to the expected inclination, the palaeolatitude calculated from the inclination places the Salinian terrane well to the south of its present position (the northward displacement is about 2 500 km). Following selective sampling to avoid an overprint from chemical remanent magnetization and to minimize viscous remanent magnetiza-

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tion and careful demagnetization (usually alternating field), Champion *et al.* (1984) also conclude that turbidites can be used successfully in palaeomagnetic investigations.

Certainly in tectonic studies where the inclination is measured in sediment and there is no independent information about possible inclination error (King 1955), it is important that a section be used that spans more than one polarity. In addition, the presence of an error in the palaeomagnetic directions can be detected most easily if the sampled beds are steeply dipping around an approximate east–west fold axis.

CONCLUSIONS

(1) Fine-grained turbidite beds in the Pico Formation exposed along Santa Paula Creek in the western Ventura Basin accurately record palaeomagnetic polarity but have a steep positive inclination and a shallow negative inclination when compared to the axial dipole field. There is also an error in the declination (biased to the east) that results from incomplete removal of a secondary component of normal magnetization that presumably was acquired in the Brunhes Normal Chron.

(2) The minerals that record the primary component of magnetization have blocking temperatures up to 450 °C. The natural remanent magnetization of almost all subsamples contains a secondary component of reverse polarity that predates folding and masks normal polarity at temperatures below about 200 °C. Compaction of the sediment, if it occurred, did not affect the alignment of the magnetic grains if the method of acquisition was depositional or post-depositional.

(3) Although there is an error in the declination of each polarity, a net clockwise rotation of about 20° is deduced for the Pico Formation north of the Oak Ridge Fault when the declination for all the sites is averaged. The rotation is similar in magnitude to that reported for the Saugus Formation in the eastern part of the Ventura Basin (Levi *et al.* 1986). The overall rate of rotation in the basin is comparable to the rate elsewhere in the Western Transverse Block as recorded in Miocene volcanic rocks and in the Monterey Formation (Hornafius *et al.* 1986).

(4) Because there are no radiometric dates or tephra layers that can be used to key the palaeomagnetic polarity in the Santa Paula Creek section to the polarity time-scale, there are several possible interpretations for the age of the section in the Pliocene.

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